



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

GENCC501

Cloud Computing Services

Consume Cloud Computing Services

Competence

RQF Level: 5

Learning Hours



50

Credits: 5

Sector: ICT and Multimedia

Trade: Computer System and Architecture

Module Type: General Module

**Curriculum: ICTCSA501 TVET CERTIFICATE V IN COMPUTER
SYSTEM AND ARCHITECTURE**

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Purpose statement	This general module describes the skills, knowledge and attitudes required to consume cloud computing services. This module is intended to prepare learners pursuing TVET Level 5 in Computer System and Architecture. Upon completion of this module, the learner will be able to Create Cloud Computing Account, manage cloud computing resources, and perform maintenance on cloud computing environment.					
Learning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	Group project and presentation	30%				
	Individual project /Work	40%				
			Summative Assessment		50%	

Elements of Competence and Performance Criteria

Elements of competence	Performance criteria
1. Create Cloud Computing Account	1.1 Cloud Computing Service Models Are Properly Selected in Line with Cloud Compliance Standards And Business Needs
	1.2 Cloud Computing Service Provider Is Appropriately Selected Based on SLA (Service Level Agreement)
	1.3 customer's account is appropriately managed based on service deployment models and user needs
2. Manage cloud computing resources	2.1 cloud resources are effectively provisioned based on workload demands
	2.2 cloud resources are properly monitored in accordance with scaling strategies for optimal performance
	2.3 cloud resources utilization is correctly optimized in accordance with resource constraints
	2.4 cloud environment security measures are properly applied based on service provider requirement and SLA
3. Perform Maintenance	3.1 Cloud resources backup strategies are appropriately applied

	based on industry standards for data resilience
	3.2 Patches are effectively applied based on the latest stable versions
	3.3 Errors are correctly fixed based on cloud computing service functionality

Knowledge	Skills	Attitude
• Description of cloud computing service models • Description of Service Level Agreement • Description of cloud computing service providers • Description of cloud resource • Description of security measures • Identification of cloud environment security Measures	• Selection of Cloud computing service providers • Management of customer's account • Monitoring cloud resources • Performing cloud resources provisioning • Fixing errors on cloud computing services • Creation of account • Backup and Disaster Recovery • Applying cloud resources utilization optimization strategies	• Being passionate • Having critical thinking • Having persistence • Being curious • Having patience • Being creative • Being Innovative • Being attentive

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Create cloud computing account. 2. Manage cloud computing resources. 3. Perform maintenance.
Learning outcome 1: Create Cloud Computing Account	Learning hours: 15
Indicative content	
● Selection of cloud computing service models ✓ Definition of key terms: ✚ Cloud computing ✚ Service models ✓ Description of cloud computing service models ✚ Infrastructure as a Service (IaaS) ✚ Platform as a Service (PaaS) ✚ Software as a Service (SaaS) ✓ Selection criteria of cloud computing service models ✚ User needs ✚ Pricing ✚ Operation ● Selection of Cloud Computing Service Provider ✓ Description of SLA ✚ Definition ✚ Components ✚ Types	

 Benefits

- ✓ Description of cloud computing service providers

 Definition

 Types (private, public, hybrid, multi cloud...)

 Case studies (pros and cons)

- ✓ Selection criteria of Cloud computing service providers

 User needs

 Pricing

 Operations

- **Management of customer's account**

- ✓ Description

 Types of customer's account

 Usage and billing

 Security and access control

- ✓ Creation of account

- ✓ Identity and Access Management (IAM)

- ✓ Backup and Disaster Recovery

Resources required for the learning outcome

Equipment

- Computer
- UPS
- projector

Materials

- Internet
- Cables

Tools	• Web browser
Facilitation techniques or Learning activity	• Practical Exercise • Individual and group work • Demonstration and Simulation • Trainer guided • Group discussion
Formative assessment methods /(CAT)	• Written assessment • Performance assessment • Project assessment • Product assessment

Learning outcome 2: Manage cloud computing resources	Learning hours: 20
Indicative content	
• Provisioning cloud resources ✓ Description of cloud resource 📌 Definition 📌 Types ✓ Perform cloud resources provisioning 📌 Assessment of Requirements 📌 Selection of cloud resource (VM, DB, Storage...) 📌 Configure cloud resources	

✓ Deprovisioning cloud resources

- **Monitoring cloud resources**

✓ Identification of cloud monitoring tools

- Digital Ocean monitoring and uptime

- AWS Cloud watch

- Microsoft Azure monitor

- Google cloud operations

✓ Description of scaling strategies

- Vertical

- Horizontal

✓ Applying cloud monitoring tools.

- **Optimization of cloud resources utilization**

✓ Identify Cloud resources utilization optimization.

- Strategies

- Best practices

✓ Apply cloud resources utilization optimization strategies.

- **Applying cloud environment security measures**

✓ Description of security measures

- Definition

- Types

- Benefits

✓ Identify cloud environment security Measures.

✓ Secure cloud environment

Resources required for the indicative content

Resources required for the indicative content	
Equipment	• Computer • UPS • Projector
Materials	• Internet • Cables
Tools	• Web browser

Facilitation techniques or Learning activity	• Practical Exercise • Individual and group work • Demonstration and Simulation • Trainer guided • Group discussion
Formative assessment methods /(CAT)	• Written assessment • Performance assessment • Project assessment • Product assessment

Learning outcome 3: Perform Maintenance	Learning hours: 15
Indicative content	
• Applying cloud resources backup strategies ✓ Description - Backup definition - Types of backups - Benefits of backup ✓ Identification of backup strategies ✓ Implement cloud backup strategies. • Applying patches ✓ Description of patches - Definition - Types	

 Working Operations

- ✓ Implementation of different patch types

- **Fixing errors in cloud computing services**

- ✓ Description of cloud errors

- Definition

- Types

- ✓ Identification of different error fixing solutions

- Automated Remediation

- Manual Troubleshooting

- Software Updates and Patching

- Scaling Resources

- Data Recovery and Backup Restoration

- ✓ Apply error fixing solutions

- ✓ Error solution Documentation

- Root cause

- Procedures

- Solution

Resources required for the indicative content

Equipment

- Computer

	• Projector
Materials	• Internet • Cables
Tools	• Web browser
Facilitation techniques or Learning activity	• Practical Exercise • Individual and group work • Demonstration and Simulation • Trainer guided • Group discussion
Formative assessment methods /(CAT)	• Written assessment • Performance assessment • Project assessment • Product assessment

References

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- NetApp. (2023, 02 10). *cloud-optimization*. Retrieved from spot.io: <https://spot.io/resources/cloud-optimization/cloud-optimization-the-4-things-you-must-optimize/>
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